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Adapting standardized patient training to improve patients' understanding and preparedness for health care encounters

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ABSTRACT

Objective: To examine the impact of standardized patient (SP) training on SPs' real-life healthcare encounters and explore whether SP training elements can be adapted to increase actual patients' understanding, communication and participation in a patient-centered care model.

Methods: Data were collected from surveys and focus groups with standardized patients and a survey of primary care physicians. Findings were used to create an educational video with pre- and post-viewing surveys of patients' understanding of engagement strategies and plans to use them in future encounters.

Results: SPs reported medical visits were more productive because of their ability to understand the visit's framework; crediting their SP training. Patients reported the video will help in planning future medical visits by providing information that increases their understanding of their role in the care process.

Conclusions: SPs' understanding of the visit and its impact on knowledge, skills and affective domains can be transferred to patients in the form of specific strategies that enhance communication and patient participation during medical visits.

Practice Implications: A brief educational intervention for patients using SPs' understanding of the medical visit may contribute to enhanced patient participation in future health care encounters and could increase patient engagement in patient-centered models of care.

1. Introduction

Patient-centered care seeks to make patients active participant in decisions about their health. In this model, the physician strives to understand the patient's perspective to provide the best care for their needs.[1,2] One challenge is that physicians' perception of patients' needs may differ from what patients think they need.[3,4] Improving bidirectional communication may help physicians and patients attain a clearer picture of the plan of care.[4].

Medical students and other health professions trainees learn and practice communication and clinical skills with standardized patients (SPs), individuals trained to act as patients who have become an integral component of teaching and assessment of communication skills for medical and health professions students. SPs help learners practice communication and clinical skills for encounters with actual patients,[5] with more than three quarters of US medical schools using SPs.[6] SPs also function as "secret shoppers" who provide feedback on physician-patient communication.[7] Data from a national curriculum inventory shows that the number of medical schools requiring SP or objective skills clinical examinations grew significantly even prior to the discontinuation of the USMLE Step 2 Clinical Skills examination,[8] with a recent study showing 84% of medical schools conducted clinical skills examinations with third- and four-year medical students using SPs.[9] A recent realist review of 19 primary studies that examined how learning with SPs teaches patient centeredness to medical learners found several mechanisms, including enhanced confidence, a sense of comfort and safety, self-reflection, awareness, and increased ability to compare, contrast and combine different perspectives, with this training enhancing learners' knowledge, skills and affective domains for engaging with patients.[10].

Several studies have shown a positive impact of SP training on individuals' SPs' experiences as actual patients.[11–13] SPs reported their medical knowledge increased, that they better understood medical histories, physical examinations and the physician's role, and that they were more discerning about aspects of the physician-patient communication. [11,12]. SPs also were less concerned about their health encounters [11,12] and felt more in control of their health. [13] Finally, SPs perceived less power differential between them and medical

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professionals, seeing more equality in the patient-physician relationship.[13].

The positive results experienced by SPs may be relevant to the patient-centered model of care, which focuses on sharing power and responsibility among physicians and patients.[14] A review of 80 studies confirmed these concepts and added coordination of care.[15] While the patient-centered model promotes physicians inviting patients' participation, the concept does not come with recommendations for how patients should do this.

Our study explored 1) whether SP training increased SPs' preparedness to actively participate in their own health care visits and the specific activities they used; 2) whether physicians thought engagement behaviors used by SPs would be useful in actual patient encounters; and 3) whether this information could be transferred to actual patients in the form of specific strategies for the medical encounter to enhance their understanding of and participation in their care.

2. Methods

The setting for the study is the Frank H. Netter MD School of Medicine at Quinnipiac University in North Haven Connecticut (QU Netter). The QU Netter Standardized Patient and Assessment Center teaches 380 medical students and students from other health professions, with communication and clinical skills teaching provided by 70 SPs.

We conducted a mixed-methods study in 3 phases over 25 months. In phase 1, we used a survey and focus groups to confirm that SPs at our school had similarly positive perceptions of their understanding of and participation in medical encounters as found in prior studies, along with SP reports of the knowledge and strategies that contributed to this. For phase 2, we used this information to create a survey of physicians to identify patient strategies they found helpful in the medical encounter. In phase 3, we used the findings from the prior phases to develop a video targeted to actual patients. It highlighted characteristics of patient participation in medical encounters deemed helpful by SPs and physicians. We showed the video to actual patients and collected pre- and post-viewing data on patients' perceptions of their preparedness for participation in future encounters.

For quantitative data, we report frequencies and percentages and used the chi-square test to compare demographic characteristics. We used the t-test to test for differences between pre- and post-video viewing responses for actual patients. Analyses were conducted in SPSS v28 and statistical significance was set at an alpha level of 0.05.

The study was approved by the Quinnipiac University Institutional Review Board (IRB) (IRB#05322).

2.1. Phase 1

In April 2021, we sent an anonymous survey to 70 QU Netter SPs. The questionnaire encompassed questions about SPs' perceptions on the impact of their training and work on their real-life medical encounters. Questions were adapted from prior published work, particularly Wallach and colleagues' quantitative study of SPs' perceptions about their own health care.[11] The survey used yes/no and forced choice questions and open-ended questions. We aggregated data to gain an initial sense of SPs' Perception of the impact of their training and work.

To deepen our understanding of SP perceptions and to collect data on aspects of SP training that could be adapted to education for actual patients, we conducted focus groups with a sample of 17 SPs in May 2021. We selected SPs to ensure a diverse population based on age, gender, race and years of SP experience. SPs consented to participate and were assured confidentiality. Focus groups lasted 1 h, were conducted virtually in groups of 4 to 5, and were led by 2 members of the study team (GC and 1 SP, who did not participate in the study and assisted with the focus groups). Sessions were recorded and transcribed.

2.2. Phase 2

In June 2021, we examined whether SPs' perceptions of the positive impact of their training and experience were shared by physicians. Using the themes identified by the SPs in Phase 1, we developed a survey to ask primary care physicians about their perceptions whether identified by SPs contributed to patient participation in the medical encounter. The questionnaire used a Likert scale (not helpful (1) to most helpful (5)), along with open-ended questions. Physicians selected for the study are affiliated with the QU Netter medical school and their practices serve as rotation sites for its medical students.

2.3. Phase 3

In June 2023, we used the participation strategies deemed most helpful by SPs and physicians and condensed relevant SP training materials into a concise educational framework based on our SP training. It breaks the medical visit down into three parts: a beginning (the interview); a middle (the exam); and an end (communication around the plan of care). We identified strategies for each part to help patients be more active participants in the encounter.

We created an 8-minute video, with an emphasis on the 3 parts of the medical visit, strategies that contribute to enhanced participation in the encounter and how patients could apply them in their care. We developed a pre- and post-survey, which examined patients' preparation to engage in 10 communication and participation strategies prior to and after viewing the video. The post-survey prefaced the questions about the 10 strategies with "After watching the video, how likely are you to [use the particular strategy]." The post-survey also asked 1) whether patients found the video relevant and helpful; 2) whether they intended to communicate with their physician using the strategies in the video; and 3) which strategies they considered the most useful. The complete survey materials included a cover letter and an informed consent document. The survey was hosted on Qualtrics and the survey link was posted along with the video.

We recruited individuals who were not active or retired health care professionals. We excluded minors and individuals with cognitive impairment. We used 2 approaches to recruit participants. We collaborated with a local senior living community and posted the video and the pre-and post-viewing surveys on the QU Netter Standardized Patient and Assessment Center website, which is accessible to the public.

3. Results

3.1. Phase 1: SP perceptions of impact of training on preparedness for health care encounters

SP participants included 9 females and 8 males between the ages of 29 and 79, who had served as an SP between 2 and 7 years. For all 9 questions in the SP survey that assessed communication and understanding during medical encounters, more than 55% of the respondents *agreed* or *strongly agreed* that their preparedness had increased since becoming an SP (See Fig. 1).

Our results showed that the aspects of SP training that had the largest impact on how SPs interact with their care providers encompassed: 1) understanding the medical interview (88%); 2) understanding the physical exam (90%); 3) understanding physician communication (90%); and 4) enhanced ability to assess the quality of the medical visit (98%). We attribute the perception that they feel they are better able to judge the quality of a visit based on the extensive assessment training SPs receive on the use of the Medical Interview Rating Scale (MIRS)[16, 17] and other checklists based on best practices[18] for providing feedback to medical students and health professions learners.

To analyze the focus group data, we used the deductive 'attribute coding' approach described by Miles et al.[19] This allowed us to organize the focus group data into distinct categories and assigned a

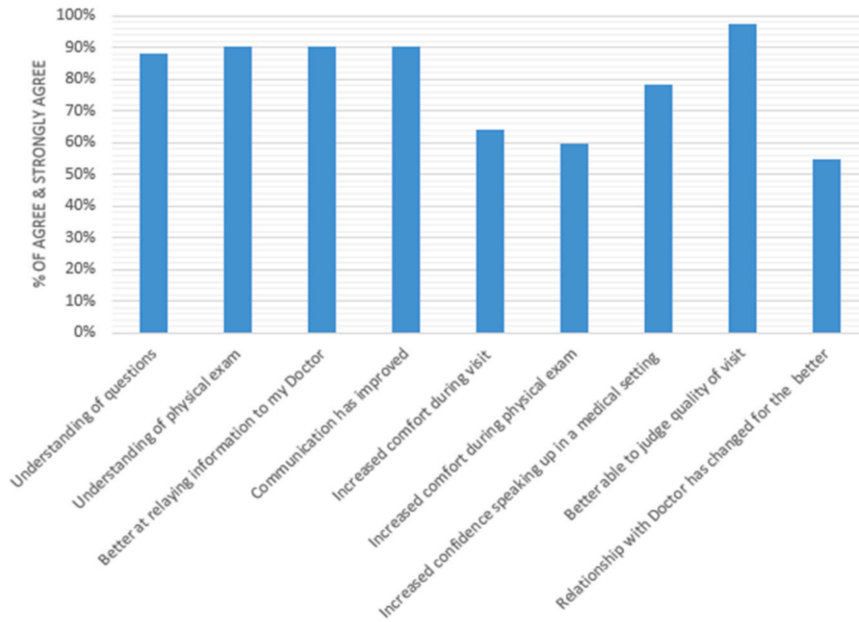

Fig. 1. *:* SP Reports of Changes in Their Actual Medical Encounters Since Becoming an SP.

unique code to each category. Codes are used to identify and analyze patterns and themes in the data, by linking codes with specific concepts or activities mentioned by the SPs during the focus groups, such as "engaging in dialogue with their physician." If a concept or activity was mentioned 10 or more times, it was considered significant. For example, being engaged in the encounter/sharing more information was mentioned 30 times, and partnering with the physician was mentioned 17 times. Using this approach, we identified 7 themes SPs felt contributed to active participation in the medical encounter. The focus group findings validated 4 findings from the SP survey, including: 1) enhanced understanding of the patient interview and the physical exam; 2) engaging in dialogue with the physician; 3) asking clarifying questions; and 4) SPs' perceptions they are better prepared to judge the quality of their care encounters. They added 3 new concepts: 1) SPs' awareness of time constraints in encounters; 2) SPs' advance preparation for their medical visits; and 3) participants sharing their role as an SP with their physician.

3.2. Phase 2: Physician perceptions of helpful patient participation strategies

In June 2021, we distributed the survey to 140 primary care physicians who serve as preceptors for QU Netter's longitudinal clinical immersion experience for Year 1 and 2 students. We received 40 responses (29% response rate).

Fig. 2 shows the 6 patient strategies physicians found most helpful: 1) patients knowing the details of their symptoms (94%); 2) patients knowing the timeline of their symptoms (100%); 3) patients recalling

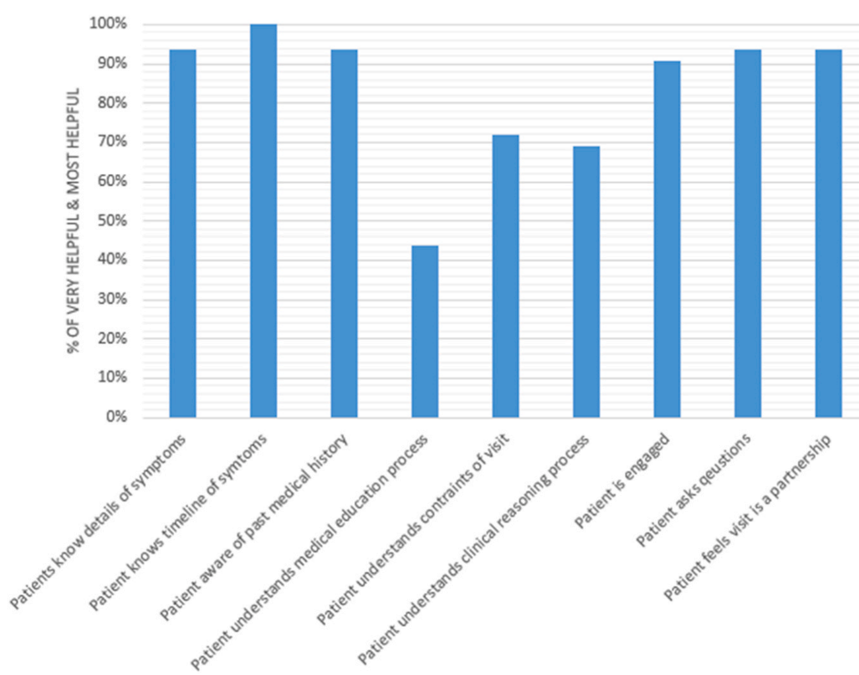

Fig. 2. Physicians Perceptions of Patient Participation Strategies that are Helpful in the Medical Encounter.

their past medical history (94%); 4) patients being engaged in their visit (91%); 5) patients asking questions to ensure understanding (94%); and 6) patients perceiving a partnership with their physician (94%).

3.3. Phase 3: Video intervention and the impact on patient perceptions of their preparedness for the medical encounter

Our pre-intervention survey was completed by 46 individuals. The majority were White (89%), female (57%), were 65 years old or older (51%), had a college degree (76%) and reported an annual income of $75,000 and above (55%). While we lack specific data on the numbers recruited through the two sites where we posted posting the patient video and survey, nearly 50% of participants were age 64 or younger, suggesting that our outreach to the senior living community and posting the video on the SP Center's web site both contributed participants. Participants had been instructed to complete the pre-survey, review of the video and the post-survey in a single setting. No reminders were sent to survey participants.

Thirty participants (65%) completed the post-survey, with the majority completing it a short time after completing the pre-survey. There were no significant differences in demographic composition between pre- and post-survey respondents. Detailed information on participants' demographics is provided as online supplemental material.

The Table 1 shows the mean response on the pre- and post-surveys of participants' use of the various participation strategies, using a scale of 0 (never) to 10 (all the time). For every strategy, there was an increase between the pre- and the post-survey and for most items the change was statistically significant ($p < .05$). When a strategy did not reach significance, it was usually due to a ceiling effect with participants reporting a high level of use prior to viewing the video. Fig. 3 shows the mean response across all questions in the pre- and post-viewing surveys and the increase in self-reported planned use of the strategies in future encounters following review of the video.

We received 29 responses to the open-ended post-survey question about what participants found most helpful from the video. Elements mentioned most frequently were learning about the 3 parts of the medical visit (62%), followed by learning the importance of preparing for the medical visit (21%).

4. Discussion and conclusion

4.1. Discussion

We adapted information used in SP training to create a patient education video, with a post-viewing survey focusing on the impact of patient understanding of their medical encounter and their plans for enhanced participation and engagement in their future healthcare.

A key finding is that patients reported they would use information in the video and that it would enhance their participation in future medical encounters. The Oxford dictionary defines, "participation" as "the act of taking part in an activity or event; the origin is the Latin word "participare," which means sharing.[20] Strategies identified in our study all relate to patient participation. This is important, because many accepted models of patient-centered care have not addressed the patient's role in what is intended to be a participatory approach. The 2001 Institute of Medicine (IOM) landmark report, *Crossing the Quality Chasm,* emphasized care providers should be "respectful of and responsive to individual patient preferences, needs, and values."[21] At the same time, the IOM report was targeted to physicians and other health professionals, with no sections of the document offering guidance to equip patients for their role in this new system.

Other research has focused on patient participation, with a focus on patient decision-making and health behavior change in chronic disease management. A review of research on engaging patients in decision-making focused primarily on tools used by physicians, including the "5 As" (Ask, Advise, Assess, Assist and Arrange for follow-up), the "5 Rs" for patient behavior change (discussing Relevance, Risks, Rewards, addressing Roadblocks and using Repetition), [22] yet many practical recommendations have not addressed the patients' role. A systematic review of 48 studies of patient engagement identified 3 levels of engagement: consultation, moving to involvement, and culminating in shared leadership with the physician.[23] This echoes findings from our study, with SPs perceiving greater equality in the relationship with their physician and primary care physicians reporting partnership with patients as a helpful attribute. The review identified strategies for enhancing patient participation, including educating patients to set the visit agenda and increase their confidence in their role in the encounter.[23] Only one-fourth of studies in the review addressed patients' experiences of the engagement process, with a limited number of these focusing on patient perceptions of practical strategies.[23] Prior research found patient engagement is an important factor in the success of care, with engaged patients living longer than patients who received similar treatment but were not engaged.[24,25].

Limitations of our study include that SP participants were selected from a single site and their perspective reflects the training at our institution. The response rate to our patient survey was small and our sample lacked diversity and was more highly educated than the general public. A key limitation is that we were not able to collect data on patients' implementation of our strategies or the impact of this intervention on patients' actual participation in their care. In a future project, we plan to use the findings from the current study with a broader sample of patients to examine how they would implement~~ed~~ the participation strategies suggested in the video in their medical encounters, along with their perception of the most helpful approaches for actual medical encounters, and how use of these strategies would affect their actual participation in their care.

By exploring how an adaptation of SP training may affect actual patients' capacity for patient-physician communication, this paper adds to the literature on patient-centered communication, with the potential to guide future investigative work in this important yet relatively underexamined area.

4.2. Conclusion

Having been trained and working as a SP has positive effects on SPs' real-life patient encounters. We assessed SPs' and physicians' perception of helpful aspects of this training and adapted key findings to develop a brief educational intervention for actual patients. Patients reported our brief educational intervention offered helpful strategies for use in future health care encounters.

4.3. Practice Implications

Elements of SP training that can be adapted to patient education include planning for the medical visit, engaging in conversation with the

Table 1
Mean (SD) of Question Response at Pre- and Post-Survey.

Question	Pre-Survey	Post-Survey	P-Value
Plan and review main concerns before visit	7.2 (2.8)	8.3 (2.2)	.062
Review medical history before visit	5.0 (3.5)	7.7 (2.0)	$< .001$
Review social history before visit	4.5 (3.3)	6.6 (3.4)	.010
Review family medical history before visit	3.3 (3.3)	6.0 (3.2)	.001
Prepare questions for doctor	7.4 (2.8)	8.7 (1.9)	.031
Provide detailed answers to the physician's questions	8.6 (2.1)	9.3 (1.0)	.103
Share thoughts about your health with the physician	8.4 (2.4)	9.3 (1.3)	.071
Ask the physician clarifying questions	8.3 (2.2)	9.2 (1.2)	.038
Contribute thoughts related to the proposed plan of care	7.9 (2.3)	9.3 (0.8)	.002
Seek to fully understand the plan of care	8.2 (1.9)	9.0 (1.2)	.026

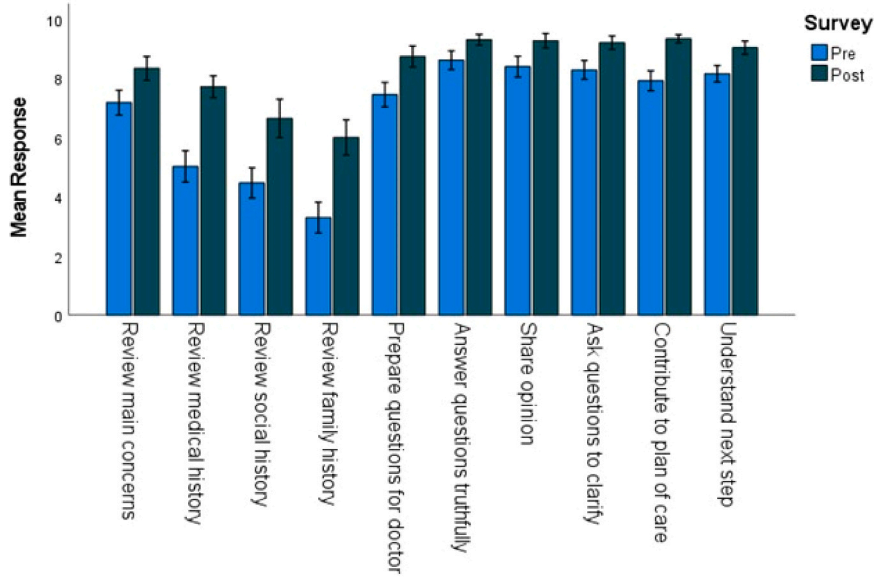

Fig. 3. Mean (± SE) Response by Survey Question at Pre- and Post-Survey.

physicians, being forthcoming with information and seeing their role in the healthcare encounter is that of a partner. These interventions may contribute to patients who are better prepared for care in a patient-centered model, allowing them to be active participants in their care.

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CRediT authorship contribution statement

Mattel] Kanevsky: Writing – review & editing, Investigation, Data curation. **Gabbriel Ceccolini:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ingrid Philibert:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Formal analysis, Data curation, Conceptualization. **Richard Feinn:** Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

All authors are at are at the Frank H. Netter MD School of Medicine at Quinnipiac University. None of the 4 co-authors (Gabbriel Ceccolini, Mattel Rachel Kanevsky, Richard Feinn, Ingrid Philibert) declare any conflicts of interest or competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.pec.2024.108276.

References

[1] Epstein RM, Peters E. Beyond information: exploring patients' preferences. JAMA 2009;302:195–7.
[2] Guyatt G, Montori V, Devereaux PJ, Schünemann H, Bhandari M. Patients at the center: in our practice, and in our use of language. ACP J Club 2004;140(1):A11–2.
[3] Street Jr RL, Haidet P. How well do doctors know their patients? Factors affecting physician understanding of patients' health beliefs. J Gen Intern Med 2011;26:21–7.
[4] Kennedy BM, Rehman M, Johnson WD, Magee MB, Leonard R, Katzmarzyk PT. Healthcare providers versus patients' understanding of health beliefs and values. Patient Exp J 2017;4:29–37.
[5] Isaksson J, Krabbe J, Ramklint M. Medical students' experiences of working with simulated patients in challenging communication training. Adv Simul (Lond) 2022;7:32.
[6] Barzansky B, Etzel SI. Educational programs in US medical schools, 2003-2004. JAMA 2004;292:1025–31.
[7] Fiscella K, Franks P, Srinivasan M, Kravitz RL, Epstein R. Ratings of physician communication by real and standardized patients. Ann Fam Med 2007;5:151–8.
[8] SP/OSCE final examinations at US medical schools. Association of American Medical Colleges. https://www.aamc.org/initiatives/cir/406426/9.html, Accessed October 2, 2023.
[9] Phillips A, Hauer KE, Chen HC, Wray A, Watanaskul S, Boscardin CK. The consequences of step 2 clinical skills examination discontinuation for medical schools and sustainability plans for clinical skills assessment. Acad Med 2023;98:717–22.
[10] Grau Canét-Wittkampf C, Diemers A, Van den Bogerd K, Schönrock-Adema J, Damoiseaux R, Zwart D, Jaarsma D, Mol S, Bombeke K, de Groot E. Learning patient-centredness with simulated/standardized patients: a realist review: BEME Guide No. 68. Med Teach 2023;45(4):347–59.
[11] Wallach PM, Elnick M, Bognar B, et al. Standardized patients' perceptions about their own health care. Teach Learn Med 2001;13:227–31.
[12] Simmenroth-Nayda A, Marx G, Lorkowski T, Himmel W. Working as simulated patient has effects on real patient life - preliminary insights from a qualitative study. GMS J Med Educ 2016. May 17;33:Doc42.
[13] Boerjan M, Boone F, Anthierens S, van Weel-Baumgarten E, Deveugele M. The impact of repeated simulation on health and healthcare perceptions of simulated patients. Patient Educ Couns 2008;73:22–7.
[14] Mead N, Bower P. Patient-centredness: a conceptual framework and review of the empirical literature. Soc Sci Med 2000;51:1087–110.
[15] Langberg EM, Dyhr L, Davidsen AS. Development of the concept of patient-centredness - a systematic review. Patient Educ Couns 2019;102:1228–36.
[16] The Master Interview Rating Scale, Copyrighted to Theresa A. Thomas Professional Skills Teaching & Assessment Center, Eastern Virginia Medical School, version updated February 2001.
[17] Talwalkar JS, Fortin AH, Morrison LJ, Kliger A, Rosenthal DI, Murtha T, Ellman MS. An advanced communication skills workshop using standardized patients for senior medical students. MedEdPORTAL 2021;17:11163.
[18] Gerzina H.A., Stovsky E. Standardized Patient Assessment Of Learners In Medical Simulation. [Updated 2023 Jul 24]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK546672/.
[19] Miles MB, Huberman MA, Saldaña J. Qualitative Data analysis: A Methods Sourcebook. Sage Publications; 2020.

[20] Oxford Dictionaries. 2023 Oxford University Press, n.d. Web, https://premium-oxforddictionaries-com.libproxy.ucl.ac.uk/, accessed September 16, 2023.

[21] Institute of Medicine (US) *Committee on Quality of Health Care in America. Crossing the Quality Chasm: A New Health System for the 21st Century*. Washington (DC): National Academies Press (US); 2001.

[22] Krist AH, Tong ST, Aycock RA, Longo DR. Engaging patients in decision-making and behavior change to promote prevention. Stud Health Technol Inf 2017;240:284–302.

[23] Bombard Y, Baker GR, Orlando E, Fancott C, Bhatia P, Casalino S, Onate K, Denis JL, Pomey MP. Engaging patients to improve quality of care: a systematic review. Implement Sci 2018;13:98.

[24] Darkins A, Kendall S, Edmonson E, Young Stressel PM. Reduced cost and mortality using home telehealth to promote self-management of complex chronic conditions: a retrospective matched cohort study of 4,999 veteran patients. Telemed J E Health 2015;21:70–6.

[25] McCarthy D., Mueller K. and Wren J., *The Commonwealth Fund Case Study. Kaiser Permanente: bridging the quality divide with integrated practices, group accountability, and health information technology, 2009*. Available from: http://www.commonwealthfund.org/∼/ media/files/publications/case-study/2009/jun/1278_mccarthy_kaiser_case_study_624_update.pdf, accessed September 16, 2023.